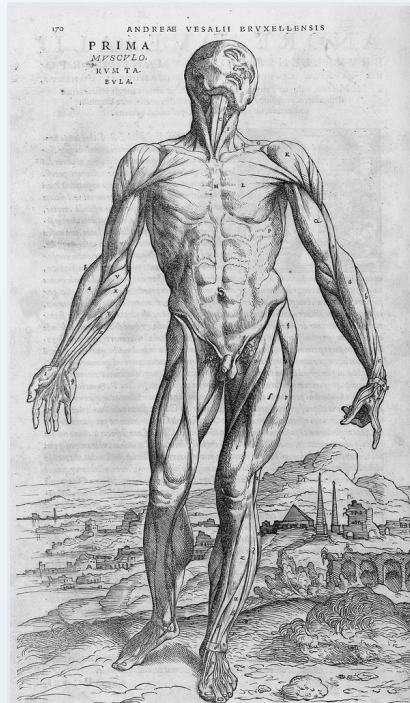


RENAISSANCE

Paper 1: Medicine Through Time with the Western Front



COURSE TEXTBOOK

Enquiry: How has medicine changed from c1250 to the modern day?

Contents

Lesson 1: KT2.1: Ideas About Cause & The Scientific Method
(Sydenham & Royal Society)

Lesson 2: KT2.2: Did treatments improve during the
Renaissance?

Lesson 3: KT2.3: How significant were William Harvey and the
Great Plague?

L1: KT2.1: Ideas About Cause & The Scientific Method (Sydenham & Royal Society)

LEARNING OBJECTIVES

- ☐ Explain how Thomas Sydenham's scientific approach to observation changed the diagnosis of disease.
- ☐ Assess the role of the printing press and the Royal Society in communicating and spreading scientific ideas.

The Medical Renaissance was ignited by an intellectual rebellion against ancient dogma. Humanism encouraged scholars to believe their own eyes rather than blindly trusting 1,400-year-old manuscripts. When young Andreas Vesalius climbed the dissection table in Padua and Thomas Sydenham sat by sickbeds in London, they transformed anatomy and diagnosis. Yet, behind this scientific triumph lay a haunting limitation: disproving Galen did not provide a single new cure, leaving ordinary citizens clinging to the Four Humours and miasma.

Did you know?

- 1676: The year Thomas Sydenham published his groundbreaking textbook *Observationes Medicae*, revolutionizing medical diagnosis.
- c.1440: The approximate year Johannes Gutenberg invented the printing press, allowing medical ideas to spread without Church censorship.
- 1665: The year the Royal Society first published *Philosophical Transactions*, the world's first scientific journal for sharing discoveries.

Recall & Retrieval

- | | |
|--|--|
| 1. In what year did the Black Death arrive in England? | 2. Who were the 'flagellants'? |
| 3. State one common medieval preventative measure for the Black Death. | 4. What are the four humours? |
| 5. What was 'miasma'? | 6. Which Roman physician developed the Theory of Opposites? |
| 7. How did medieval hospitals differ from modern ones? | 8. What was the purpose of 'bleeding' (phlebotomy) in medieval medicine? |
| 9. What was the role of a medieval apothecary? | 10. How did the medieval Church explain the cause of disease? |

Vocabulary Check

The Odd One Out: Circle *ONE* term that does not belong with the others. Explain your reasoning: What historical connection links the other terms, and why is your chosen term different?

Humanism | Nullius in Verba | Animalcules | Royal Society | Andreas Vesalius | Printing Press

Teacher Model / Pupil Reasoning:

Source S1 : Frontispiece to De Humani Corporis Fabrica (1543)



The revolutionary frontispiece to De Humani Corporis Fabrica (1543), showing Andreas Vesalius conducting a public human dissection in Padua.

Source S2 : Vesalius' 'Muscle Men' Anatomical Woodcut (1543)

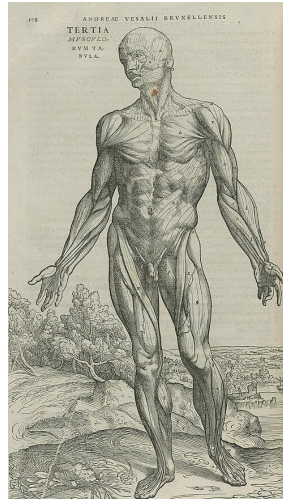


Plate from Book II of De Fabrica, depicting a flayed human figure displaying superficial musculature against an authentic Italian landscape.

Source S3 : Early Gutenberg-Style Movable Type Printing Press (16th c.)



A Renaissance printing house showing compositors setting movable metal type and pressmen operating the heavy wooden screw press.

[1.1] The Stinking Alleys of Tudor London: Why Medieval Dogma Refused to Die [1.1] For over a millennium, European medicine lived under the unquestioned authority of ancient Rome. Monks in Catholic monasteries painstakingly hand-copied the treatises of Galen and Hippocrates, treating their classical writings as infallible medical scripture. To challenge Galen was to risk excommunication or imprisonment. Medical students sat passively in cold university lecture halls while an elderly professor read aloud in Latin from an elevated pulpit, while an uneducated barber-surgeon sliced an animal carcass on the floor below. ### [1.2] The Humanist Awakening: Believing Your Own Eyes Over Ancient Dust [1.2] By 1500, this intellectual stagnation shattered under the force of the **Renaissance**—a cultural and philosophical rebirth driven by **Humanism**. Humanist scholars argued that human beings possessed reason and had a moral duty to directly interrogate the natural world rather than relying on ancient dogma. Why should modern physicians revere Roman texts written fourteen centuries earlier when the human body could be examined with their own eyes? Yet, in the squalid, smoke-choked alleys of Tudor London, ordinary citizens remained untouched by university philosophy. When epidemic fevers struck, they still looked to the heavens in terror, convinced that disease was God's wrath or the consequence of poisonous atmospheric vapors (*miasma*).



Source A: Source A: Frontispiece to *De Humani Corporis Fabrica* (1543)

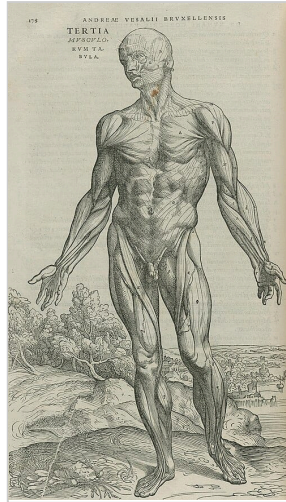
Andreas Vesalius — Pioneering Anatomist

A Renaissance physician who published "On the Fabric of the Human Body" (1543). By performing his own human dissections, he proved over 300 errors in Galen's work (e.g., proving the human jaw bone is one piece, not two).

Key actions: Conducted direct, systematic human dissections of executed criminals and closely supervised standard, printed anatomical illustrations of his findings.

Significance: Published *De Humani Corporis Fabrica* (1543). Disproved over 300 of Galen's classical errors, proving that the human jaw is a single bone, the liver has two lobes, and the septum contains no pores. Turned anatomy into an empirical, observational science.

Limitations: His anatomical breakthroughs had no immediate clinical application; knowing the exact structure of the skeleton did not help physicians cure illnesses, meaning daily medical treatments remained unchanged. Many traditional physicians initially ignored or fiercely attacked his work.



Source B: Source B: Vesalius' 'Muscle Men' Anatomical Woodcut (1543)

[2.1] The Dissection Mutiny: Slicing Human Flesh in Padua [2.1] In the winter of 1543, twenty-eight-year-old Flemish physician **Andreas Vesalius**, Professor of Surgery at the University of Padua, tore down the medieval medical hierarchy. Sweeping aside the lecturing professor and the barber-surgeon, Vesalius stood directly at the dissection table, rolled up his sleeves, and dissected human corpses of executed criminals with his own hands. What he revealed sent shockwaves through European academia: Galen had made over **300 fundamental errors**. Galen had claimed the human lower jaw was composed of two bones—Vesalius proved it was one solid bone. Galen claimed the human breastbone had seven segments—Vesalius proved it had three. Most radically, Galen asserted that blood seeped between the left and right ventricles through invisible pores in the heart's septum—Vesalius proved the septum was an impenetrable, solid muscular wall. Galen had never dissected a human being; he had dissected apes, pigs, and dogs, transposing animal anatomy onto humans. ### [2.2] The Alpine Pack-Mule Dash: Printing *De Fabrica* in Basel (1543) [2.2] Yet, Vesalius knew that revolutionary discoveries were useless if they were silenced inside a single Italian university. Medieval anatomists who questioned Galen had been dismissed or imprisoned. What transformed Vesalius's rebellion into an unstoppable European movement was a monumental collision between **individual courage, Renaissance art, and Gutenberg's printing press**. Vesalius hired master draftsmen from the Venetian workshop of the Renaissance painter **Titian** to draw 277 dynamic, mathematically precise anatomical woodcuts. He then packed these precious pear-wood printing blocks onto pack-mules, traversed the freezing alpine passes, and carried them north to the Swiss city of Basel. There, he partnered with master printer **Johannes Oporinus**. In 1543, Oporinus's presses produced *De Humani Corporis Fabrica* (*On the Fabric of the Human Body*). Thousands of identical, highly illustrated copies were bound, shipped, and distributed across European universities in weeks—far too fast for conservative church censors to intercept them.

Q1. GCSE Analytical Assessment: Evaluate both the revolutionary anatomical breakthroughs and the critical practical limitations of Vesalius.

[3.1] The Bedside Rebel: Thomas Sydenham and the Don Quixote Prescription [3.1] While Vesalius mapped the human skeleton in the 16th century, the question of what actually made living patients sick remained completely unanswered. In 17th-century London, university physicians still diagnosed illnesses from armchair theories, inspecting flasks of urine and calculating astrological horoscopes. Enter **Thomas Sydenham** (1624–1689), nicknamed the '**English Hippocrates**'. Sydenham was an outsider: a Puritan captain who had fought in the English Civil War, he despised university speculation. When an eager young medical student asked Sydenham which books he should read to become a great doctor, Sydenham bluntly replied: *"Read Don Quixote—it is a very good book. I read it still!"* To Sydenham, dry medical textbooks were pure fantasy compared to the real world of sick human beings: *"You must go to the bedside; there alone can you learn the true nature of disease."* ### [3.2] Classifying Disease Like Plants: The Fatal Blow to the Four Humours [3.2] Sitting by his patients' bedsides day after day, monitoring pulses, recording temperatures, and writing detailed clinical notes, Sydenham made a profound conceptual breakthrough. He argued that diseases were not unique, personalized imbalances of an individual's humours. Instead, diseases were **external biological entities that could be classified into distinct species**, just as botanists classify plants and flowers. In his masterwork, *Observationes Medicae* (1676), Sydenham proved that diseases had fixed, observable life cycles. He was the first physician in history to distinguish **measles from scarlet fever** as two entirely different illnesses. ### [3.3] Peruvian Bark vs Boiling Blankets: Fresh Air Over Suffocation [3.3] Crucially, Sydenham introduced rational, empirical therapies. When smallpox struck, traditional doctors locked patients in stifling, windowless rooms, piling on heavy blankets and boiling fires to 'sweat out the poison'—a practice that suffocated thousands of children. Sydenham threw open the windows, prescribed fresh air, cool drinks, and light blankets, dramatically increasing survival rates. Furthermore, he popularized the use of **Cinchona bark** (imported from Peru, containing quinine) to cure malarial 'ague' fevers, proving that specific diseases responded to specific chemical remedies.

KNOWLEDGE CHECK (Q2)

How did Sydenham's belief that diseases were like plants or animals challenge the traditional Theory of the Four Humours?

- ☐ A. It proved that diseases were caused by invisible germs.
- ☐ B. It meant diseases were separate entities with specific symptoms, rather than a personal imbalance of a patient's humours.
- ☐ C. It showed that diseases could only be cured by herbal remedies.
- ☐ D. It proved that diseases were a punishment from God.

Q3. GCSE Analytical Assessment: Evaluate both the diagnostic advancements and the practical treatment limitations of Thomas Sydenham.

Thomas Sydenham — The "English Hippocrates"

A highly influential English physician of the 17th century. He strongly advocated for the direct observation of patient symptoms and classifying diseases (like measles and scarlet fever) into distinct groups, moving away from the Theory of the Four Humours.

Key actions: Rejected book-based dogmas and spent years closely observing patients at their bedsides, systematically recording and classifying their symptoms.

Significance: Published *Observationes Medicae* (1676). Proven to be the first to classify diseases as separate species, famously distinguishing scarlet fever from measles. Popularized natural treatments, such as prescribing fresh air, rest, and cinchona bark (quinine) for malaria.

Limitations: Although he revolutionized diagnosis, he was unable to explain the biological cause of disease because Germ Theory did not exist. He still believed in miasma and held traditional views, meaning many of his therapies remained observational rather than truly curative.



Source C: Source C: Early Gutenberg-Style Movable Type Printing Press (16th c.)

[4.1] Smashing the Scribes: Gutenberg's Machine Defies Church Censorship [4.1] Invented by Johannes Gutenberg in c.1440, the movable metal type printing press took book production completely out of the hands of monastic scribes. Before printing, manuscripts took months to copy by hand, cost a small fortune, and were strictly censored by the Catholic Church. The printing press shattered this bottleneck, allowing revolutionary texts like Vesalius's *De Fabrica* (1543) and William Harvey's *De Motu Cordis* (1628) to be mass-produced and shipped across Europe before conservative church authorities could ban or burn them. ### [4.2] The Latin Luxury: Why Mass Books Left Ordinary Peasants in the Dark [4.2] Yet, this communication revolution had sharp social limits. The vast majority of printed medical treatises were published in Latin—a language understood only by wealthy scholars, aristocrats, and elite physicians. For the ordinary labourer or illiterate peasant in Tudor and Stuart England, these leather-bound university volumes might as well have been written in hieroglyphics. Commercial printers quickly discovered that cheap astrological almanacs, folk herbals, and traditional Galenic guides were far more profitable to print in English than complex scientific treatises, keeping ancient superstitions firmly alive among the public.

Q4. GCSE Analytical Assessment: Evaluate both the revolutionary impact of the printing press and its historical limitations.

[5.1] "Take Nobody's Word For It!": The Royal Charter of 1662 [5.1] Sydenham's insistence on physical observation reflected a broader institutional revolution in London. In November 1660, a group of twelve natural philosophers met at Gresham College to establish an experimental academy. In 1662, King Charles II granted them a royal charter, founding **The Royal Society**. The Society adopted the radical motto **Nullius in Verba** ("Take nobody's word for it"). They banned theological debate and declared that no scientific statement could be accepted as truth without physical experiment, mathematical calculation, or public demonstration. In 1665, the Society's first secretary, Henry Oldenburg, created Europe's first peer-reviewed scientific journal: **Philosophical Transactions**, creating an international network for sharing verified lab notes. ### [5.2] Weighing Air & Royal Giggles: Charles II and the Gentleman Philosophers [5.2] King Charles II famously found his own scientific society hilarious. The Merry Monarch would laugh heartily at the gentlemen philosophers for spending entire, solemn afternoons in stuffy rooms doing nothing more grand than carefully weighing jars of air on delicate brass scales! Yet behind the royal giggles lay an unprecedented transformation: experimental science had received the official stamp of the Crown, turning natural philosophy from a suspicious, heretical hobby into a prestigious national

enterprise. ### [5.3] The Tooth-Scraping Dutchman: Leeuwenhoek's Dancing Beasties [5.3] The most mind-boggling breakthrough came from an uneducated Dutch draper named **Antonie van Leeuwenhoek**. Leeuwenhoek ground tiny glass spheres into microscopes capable of magnifying objects up to 270 times. One morning, he used a quill to scrape the yellowish tartar from between his unbrushed teeth, mixed it with clean rainwater, and placed it under his lens. To his absolute astonishment, he saw millions of tiny microscopic organisms swimming, tumbling, and somersaulting—creatures he christened '**animalcules**' (little animals). When he wrote to the Royal Society declaring that there were more living creatures dancing on his front teeth than people in the entire Kingdom of the Netherlands, the English gentlemen gasped in wonder. Yet, despite this marvel, no one realized these microscopic 'beasties' were deadly pathogens; they were treated merely as amusing optical party tricks for another 200 years.

Q5. GCSE Analytical Assessment: Evaluate both the institutional breakthrough of the Royal Society and its practical limitations.

[6.1] The Great Renaissance Paradox: Brilliant Minds, Helpless Patients [6.1] Despite the breathtaking triumphs of Vesalius, the printing press, Sydenham, and the Royal Society, the Renaissance revealed a tragic historical paradox: **revolutionary science did not cure disease**. Knowing that the human jaw had one bone did not stop an infected wound from turning gangrenous. Knowing that diseases were distinct species did not explain *what* caused them. Microscopes existed, but Leeuwenhoek's 'animalcules' were viewed as fascinating natural curiosities rather than deadly pathogens. The true cause of disease—microscopic germs—remained undiscovered for another two hundred years until Louis Pasteur in 1861. ### [6.2] Bedside Continuity: Why Your Stuart Doctor Still Reached for Leeches [6.2] Consequently, for ordinary sick citizens, medical treatment experienced almost total continuity with the Middle Ages. When Sydenham's observation failed to break a violent fever, he still reached for his lancet to bleed the patient or prescribed poisonous calomel to purge their bowels. Conservative physicians fiercely resisted the new science; his former tutor in Paris branded Vesalius an arrogant madman, insisting that human anatomy must have deformed since Galen rather than admit Galen was wrong. For the average family in Tudor and Stuart Britain, hospital care, herbal folk remedies, and superstitious prayers remained virtually unchanged.

Pair & Share Activity

Q6. In the period c1500–c1700, was the work of individual physicians like Thomas Sydenham more important in challenging traditional medical beliefs than institutional developments like the founding of the Royal Society?

Historian's Corner: The Debate over the Pace of Change in Renaissance Medicine

Historians often debate the extent to which the Renaissance c1500–c1700 represented a genuine breakthrough or a period of stagnation in the understanding of disease causes. Traditional historians emphasize a 'Scientific Revolution,' pointing to the works of Vesalius, Harvey, and Leeuwenhoek as a rapid and complete dismantling of medieval Galenism. However, revisionist historians argue that this period was actually characterized by overwhelming continuity for the average patient. They argue that while elite physicians in organizations like the Royal Society debated new ideas, daily treatments (such as bleeding and purging to balance the humours) remained completely unchanged. Because there was no scientific proof that Leeuwenhoek's 'animalcules' actually caused disease, everyday medical

practice clung tightly to the Theory of the Four Humours and miasma for centuries after the Renaissance.

GCSE Exam Practice

Q7. Explain one way in which ideas about the causes of disease in the Renaissance period (c1500–c1700) were similar to ideas about the causes of disease in the medieval period (c1250–c1500). (4 marks)

Q8. Explain why there were changes in the way ideas about the causes of disease and illness were communicated in the period c1500–c1700. (12 marks)

L2: KT2.2: Did treatments improve during the Renaissance?

LEARNING OBJECTIVES

- ☐ Analyze the continuities and changes in Renaissance treatments and preventions, including the impact of exploration and alchemy.
- ☐ Explain how institutional changes, specifically the Dissolution of the Monasteries, altered hospital care in England.

The Medical Renaissance saw an explosion of scientific curiosity, yet this rarely translated into better everyday medical care. While pioneer anatomists fundamentally transformed medical training by proving ancient authorities wrong, the average patient still relied on traditional herbal remedies, bleeding, and local wise women. The unexpected closure of monastery hospitals forced a massive rethink in healthcare, slowly paving the way for specialised hospitals focused on actually curing the sick.

Did you know?

- 1536: The year King Henry VIII initiated the Dissolution of the Monasteries, forcing the vast majority of English hospitals to close.
- 1543: The exact year Andreas Vesalius published his revolutionary, highly accurate anatomy textbook, *On the Fabric of the Human Body*.
- 300: The approximate number of anatomical mistakes Vesalius proved Galen had made because the Roman physician had dissected animals instead of humans.

Recall & Retrieval

- | | |
|---|---|
| 1. What was the Renaissance? | 2. What invention helped spread new medical ideas during the Renaissance? |
| 3. Who was Thomas Sydenham? | 4. Who wrote 'On the Fabric of the Human Body' (1543)? |
| 5. What did Vesalius prove about Galen's anatomical ideas? | 6. In what year did the Black Death arrive in England? |
| 7. Explain the role of the apothecary in medieval medicine. | 8. Why did the influence of the Church begin to decline in the Renaissance? |
| 9. What was the Royal Society? | 10. What was 'miasma'? |

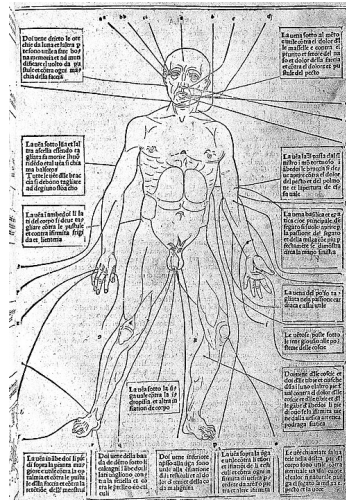
Vocabulary Check

The Golden Sentence: Choose **TWO** terms from the word bank. Write **ONE** grammatically sophisticated, historically accurate sentence connecting them using a causal conjunction (because, although, or consequently):

Transference | Iatrochemistry | Thomas Sydenham | Pharmacopoeia | Dissolution of the Monasteries
| Pest House

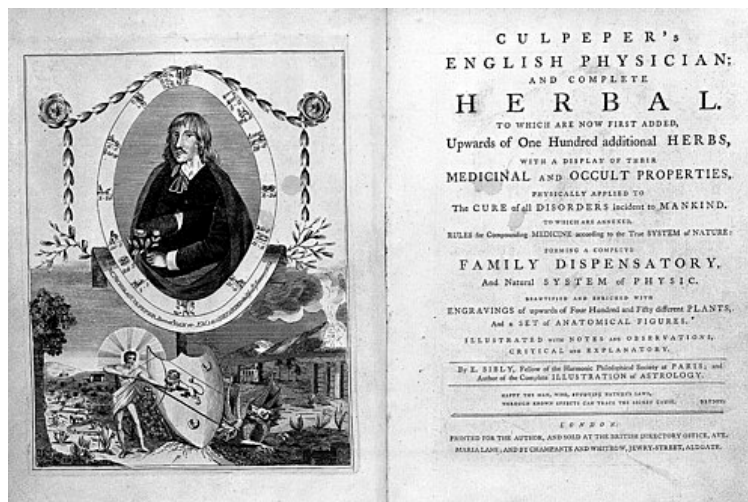
Teacher Model / Pupil Sentence:

Source S1 : 16th-Century Phlebotomy and Humoural Bloodletting Woodcut



Woodcut diagram from a 16th-century Italian medical text detailing specific vein points on the human body to be lanced for humoral bloodletting.

Source S2 : Nicholas Culpeper's English Physitian & Complete Herbal (1652)



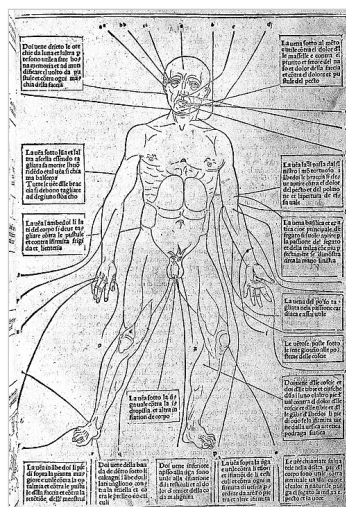
Title page of Nicholas Culpeper's *The English Physitian*, the first major medical guide to publish herbal remedies and medical knowledge in plain English.

Source S3 : Jan Steen: The Quack Doctor Selling Elixirs (c.1650)



Jan Steen satirical painting De Kwakzalver (The Quack Doctor), depicting an itinerant charlatan peddling miracle elixirs and teeth extractions on an outdoor stage.

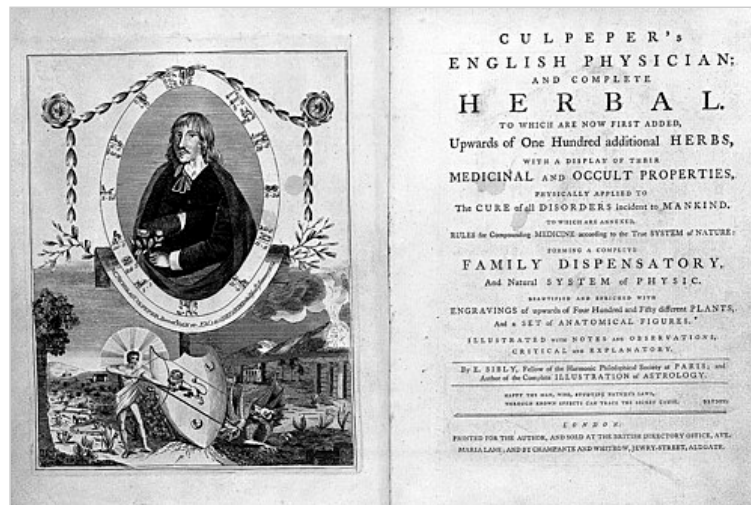
Treatments in the Renaissance Despite the brilliant discoveries in anatomy during the Renaissance, treatments for ordinary people hardly changed at all. Doctors still could not cure infectious diseases, so they continued to rely on traditional methods like bloodletting, purging, and herbal remedies. The only real changes in treatment came from the discovery of the 'New World' (the Americas). Explorers brought back new herbs and plants, such as the bark of the Cinchona tree, which was used to treat malaria, and sarsaparilla, which was used to treat syphilis. Furthermore, Thomas Sydenham, known as the 'English Hippocrates', began encouraging doctors to observe their patients' symptoms closely and treat the specific disease, rather than just treating the symptoms individually.



Source A: Source A: 16th-Century Phlebotomy and Humoural Bloodletting Woodcut

Continuity and Change in Treatment Bleeding, purging, and sweating remained the most common everyday treatments because the general public and physicians still firmly believed that an imbalance of the Four Humours caused disease. A newly popular theory was transference-the belief that an illness could be transferred to an object or animal (for example, rubbing warts with an onion, or sleeping next to a sheep to cure a fever). Global exploration brought new exotic herbal remedies from the New World (the Americas), such as the use of cinchona bark, which contains quinine, to successfully treat malaria. The growth of alchemy led to iatrochemistry (medical chemistry), popularized by the physician Paracelsus. This introduced powerful mineral-based cures, such as antimony and mercury, to forcefully purge the body of disease, which challenged traditional plant-based herbalism. Although exotic new herbs and powerful chemicals were introduced, treatment experienced massive continuity overall because doctors were still fundamentally trying to fix the body's internal balance rather than fighting germs.

Q1. GCSE Analytical Assessment: Evaluate the extent to which medical treatments improved during the Renaissance (c1500–c1700).



Source B: Source B: Nicholas Culpeper's English Physitian & Complete Herbal (1652)

Continuity and Change in Prevention People still desperately tried to avoid breathing in miasma (bad air) by carrying sweet-smelling herbs, and town councils continued to enforce the removal of rotting sewage from streets. People still followed the Regimen Sanitatis (moderation in diet and exercise), but also began using new instruments like barometers to try and link specific weather conditions to outbreaks of disease. Crucially, bathing became far less fashionable. The terrifying spread of syphilis (the Great Pox) across Europe had been linked to public bathhouses, so people instead tried to keep clean by regularly changing their linen clothes. Despite changing attitudes towards public bathing and a more scientific approach to observing the weather, prevention methods remained largely ineffective because the true microscopic causes of disease remained invisible.

KNOWLEDGE CHECK (Q2)

Why was the importation of cinchona bark from South America a more successful treatment than traditional humoral methods?

- ☐ A. It contained quinine, which actually treated the specific symptoms of malaria, rather than just attempting to rebalance humours.
- ☐ B. It was a holy relic blessed by the Pope.
- ☐ C. It killed the germs that caused malaria.
- ☐ D. It was used to draw bad humours out of the skin.

Q3. Describe the approach of Thomas Sydenham.



Source C: Source C: Jan Steen: *The Quack Doctor Selling Elixirs* (c.1650)

Medical Care: Hospitals and the Community In 1536, King Henry VIII ordered the Dissolution of the Monasteries. Because the vast majority of hospitals were run by monks and nuns on church land, the hospital system was devastated, shutting down almost all hospital beds in England. A few hospitals, such as St Bartholomew's, only survived because the citizens of London petitioned the King directly to refound it as a municipal hospital. Slowly, new charity-funded hospitals opened that actually employed physicians to focus on curing the sick, rather than just caring for their souls. Pest houses (or plague houses) began to appear; these were specialized isolation hospitals created specifically to quarantine patients suffering from highly contagious diseases. Apothecaries and surgeons also became more organised and professional during this period. In 1540, the Company of Barber-Surgeons was formed when Henry VIII merged the barbers' and surgeons' guilds, giving surgery its own formal training structure and separating it from simple haircutting for the first time. In 1617, apothecaries broke away from the Grocers' Company to form their own Society of Apothecaries, which trained and licensed its members to legally prepare and sell medicines, helping to squeeze out untrained 'quack' healers. Despite these changes, the vast majority of sick people continued to be cared for at home by female family members mixing cheap herbal remedies. The closure of the monasteries temporarily devastated the hospital system, but ultimately triggered a vital shift towards hospitals becoming places of practical medical treatment rather than religious rest homes.

Q4. Explain why treatments for ordinary people hardly changed during the Renaissance.

Q5. Describe one way apothecaries or surgeons became more professional during the Renaissance.

Medical Training and the Work of Vesalius Physicians still trained at universities, but dissection was finally legalised and became a central, highly fashionable part of their education. Students learned using highly detailed printed fugitive sheets (anatomical drawings with liftable paper layers). The most famous anatomist of the period was Andreas Vesalius, a lecturer in surgery at the University of Padua. In 1543, Vesalius published his groundbreaking, heavily illustrated anatomy book: *On the Fabric of the Human Body*. By performing dissections on executed criminals, Vesalius proved that Galen had made over 300 mistakes. For example, he proved the human jawbone is one solid bone, not two; that the breastbone has three segments, not seven; and that blood does not flow through invisible pores in the heart's septum. Vesalius revolutionized medical training because he proved that the ancient texts were flawed, actively encouraging generations of doctors to perform their own dissections and believe their own eyes rather than blindly trusting tradition.

Q6. To what extent did the 'New World' improve medical treatments?

Pair & Share Activity

Q7. In the period c1500–c1700, did the introduction of global herbal remedies and chemical cures represent a more significant improvement in treatment than the gradual reforms made to the training of physicians?

Historian's Corner: The Debate over Renaissance Progress: Breakthrough or Backstep?

Historians actively debate whether the Renaissance represented genuine progress in patient care or a period of regression. Traditionalist accounts often celebrated the 'rebirth' of medical science, highlighting how the shift towards chemical cures and Vesalian anatomy laid the foundations of modern clinical practice. However, revisionist social historians argue that for the average sick person, the Renaissance was actually a time of decline in care. They point out that Henry VIII's closure of the monasteries in 1536 destroyed the medieval hospital safety net, leaving thousands of poor patients without any institutional care. Furthermore, they highlight that some new treatments, such as using highly toxic mercury to treat syphilis or smoking tobacco as a 'cure-all' defense against plague, actually did far more physical harm than medieval herbal remedies ever did.

GCSE Exam Practice

Q8. Explain why there was little change in medical treatments during the Renaissance period (c1500–c1700), despite new anatomical discoveries. (12 marks)

Q9. Explain one way in which medical treatments in the Renaissance period (c1500–c1700) were similar to medical treatments in the medieval period (c1250–c1500). (4 marks)

L3: KT2.3: How significant were William Harvey and the Great Plague?

LEARNING OBJECTIVES

- ☐ Explain the significance of William Harvey's discovery and the reasons for its limited short-term impact.
- ☐ Compare the government and public responses to the 1665 Great Plague with the 1348 Black Death.

By the 17th century, medical knowledge was advancing, but practical treatment lagged far behind. William Harvey's brilliant discovery of blood circulation proved ancient theories wrong and laid the foundations for modern physiology, though it saved no lives at the time. Meanwhile, the devastating Great Plague of 1665 demonstrated that without an understanding of germs, even organized government action was powerless to stop a catastrophic epidemic.

Did you know?

- 1628: The year William Harvey published his groundbreaking book proving the circulation of the blood.
- 100,000: The approximate number of Londoners who died during the Great Plague of 1665 (one-fifth of the city's population).
- 28: The number of days an infected family was locked inside their house under the strict new quarantine rules.

Recall & Retrieval

- | | |
|---|--|
| 1. Who wrote 'On the Fabric of the Human Body' (1543)? | 2. What did Vesalius prove about Galen's anatomical ideas? |
| 3. What was 'transference'? | 4. What was the Royal Society? |
| 5. Why did the influence of the Church begin to decline in the Renaissance? | 6. In what year did the Great Plague hit London? |
| 7. How did local mayors try to prevent the spread of the Black Death in 1348? | 8. What was the Theory of the Four Humours? |
| 9. What was the purpose of 'bleeding' (phlebotomy) in medieval medicine? | 10. What was the role of a medieval apothecary? |

Vocabulary Check

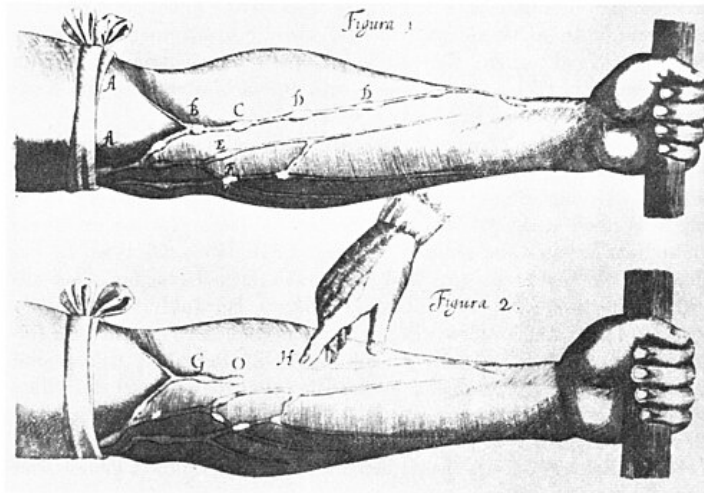
Conceptual Classification: Categorise the terms from the word bank into the two historical boxes below:

Circulation | William Harvey | Great Plague | Bills of Mortality | Searchers | Plague Doctor

Power, Governance & Warfare

Economy, Trade & Society

Source S1 : William Harvey's Arm Experiment with Valves in Veins (1628)



Engravings from Harvey's De Motu Cordis demonstrating how one-way valves in the arm veins permit blood to flow only toward the heart.

Source S2 : The London Bill of Mortality for the Great Plague (1665)

[illegible]

General Bill of Mortality for the City of London recording 7,165 plague deaths in a single week at the height of the epidemic in September 1665.

Source S3 : Engraving of 'Doktor Schnabel von Rom' (The Plague Doctor, 1656)



The iconic plague doctor costume: heavy waxed leather coat, spectacles, a long bird-like beak filled with aromatics, and a wooden cane to examine corpses.

William Harvey and the Great Plague In 1628, William Harvey made one of the most important discoveries in medical history: he proved that blood circulates around the body, pumped by the heart. He completely destroyed Galen's theory that blood was constantly being made in the liver and burned up by the muscles. Harvey proved this by dissecting cold-blooded animals and experimenting on human veins. However, in 1665, the Great Plague struck London, proving that while anatomical knowledge had improved, prevention of disease had not. People still blamed miasma and the planets, though local governments did take stronger action this time. The Mayor of London ordered victims to be locked in their houses with a red cross painted on the door, and banned public gatherings. Yet, just like in 1348, the true cause (fleas on rats) remained unknown.

William Harvey — Royal Physician

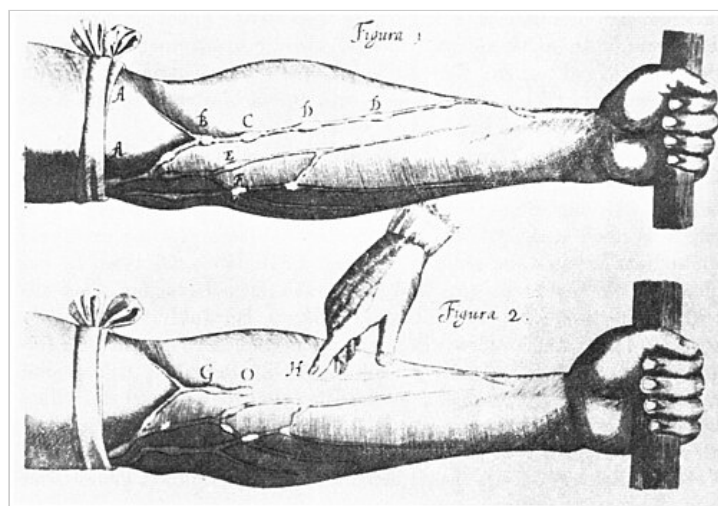
An English physician who discovered the full circulation of blood in the human body. By dissecting cold-blooded animals and experimenting with ligatures, he proved the heart acts as a pump, disproving Galen's theory that blood was constantly manufactured in the liver.

Key actions: Dissected cold-blooded animals to observe slow heartbeats, and used a system of rods and ligatures on human arms to prove that blood flows only one way through veins.

Significance: Published *De Motu Cordis* (1628). Proved that the heart acts as a muscular pump circulating blood around the body, disproving Galen's theory that blood was constantly manufactured by the liver and absorbed by the tissues.

Limitations: His discovery had no direct clinical treatment value at the time, as knowing blood circulated could not cure disease. He lacked microscopes powerful enough to prove the existence of capillaries (which connect arteries and veins; discovered by Malpighi in 1661). Many contemporary doctors rejected his findings, and his ideas did not appear in English university textbooks until decades later.

Q1. GCSE Analytical Assessment: Evaluate both the revolutionary physiology of William Harvey and its total lack of practical treatment impact.



Source A: Source A: William Harvey's Arm Experiment with Valves in Veins (1628)

William Harvey and the Circulation of the Blood William Harvey was an English doctor who studied at the famous medical university in Padua, Italy, before eventually becoming the personal royal physician to King Charles I. In 1628, he published his groundbreaking book, *An Anatomical Account of the Motion of*

the Heart and Blood in Animals. By using mechanical pump calculations and dissecting cold-blooded animals (whose slower heartbeats allowed him to observe muscle movement), he proved that blood circulates around the body in a one-way system, and that the heart acts as a muscular pump. He proved that the ancient Roman physician Galen was wrong when he claimed that blood was constantly manufactured in the liver and absorbed as fuel by the body. Harvey's work was a massive scientific breakthrough because it proved that traditional ancient texts were flawed and laid the vital foundations for modern physiology and future developments in surgery and blood transfusions.

Q2. Identify what William Harvey proved about the blood.

The Limits of Harvey's Discovery Despite his brilliance, Harvey's discovery had almost no immediate impact on everyday medical treatment because understanding how blood circulates did not help a doctor cure a fever or treat a disease in 1650. Many traditional doctors initially rejected his ideas because they went against Galen, and some even openly criticised him. It took until 1673 for his theories to be widely taught in universities. Furthermore, he could not explain how blood moved from arteries to veins because capillaries were invisible to the naked eye. This gap between brilliant new science and unchanged everyday treatment is a useful revision idea to remember (some teachers nickname it the 'Stagnation Paradox' as a memory aid, though it is not a term you need to quote in an exam): Vesalius and Harvey both had almost zero immediate impact on bedside treatment. Knowing that blood circulated did not give physicians new drugs, meaning patients still demanded traditional humoral bleeding and purging.

KNOWLEDGE CHECK (Q3)

What was the main reason why Harvey's discovery of blood circulation did not immediately change medical treatments?

- ☐ A. Because he refused to publish his findings in a book.
- ☐ B. Because the Church banned his book and executed him.
- ☐ C. Because knowing how the blood circulated did not provide doctors with any new drugs or cures for sick patients.
- ☐ D. Because he still believed the liver manufactured blood.

Q4. GCSE Analytical Assessment: Evaluate the advancements in government organization against the ongoing medical ignorance of the 1665 plague.

A general Bill for this present year, ending the 1st of December 1665, according to the Report made to the S^{ts} C^{ts} and C^{ts} Mortality.

By the Company of Parish Clerks of London, &c.

Parish	Male	Female	Total
St. Andrew's	12	10	22
St. Dunstons	15	12	27
St. Giles	18	15	33
St. Martin	20	18	38
St. Michael	22	20	42
St. Nicholas	25	22	47
St. Peter	28	25	53
St. Paul	30	28	58
St. James	32	30	62
St. John	35	32	67
St. Mary	38	35	73
St. Anne	40	38	78
St. Elizabeth	42	40	82
St. George	45	42	87
St. David	48	45	93
St. Thomas	50	48	98
St. Richard	52	50	102
St. Edmund	55	52	107
St. Andrew	58	55	113
St. Dunstons	60	58	118
St. Giles	62	60	122
St. Martin	65	62	127
St. Michael	68	65	133
St. Nicholas	70	68	138
St. Peter	72	70	142
St. Paul	75	72	147
St. James	78	75	153
St. John	80	78	158
St. Mary	82	80	162
St. Anne	85	82	167
St. Elizabeth	88	85	173
St. George	90	88	178
St. David	92	90	182
St. Thomas	95	92	187
St. Richard	98	95	193
St. Edmund	100	98	198
St. Andrew	102	100	202
St. Dunstons	105	102	207
St. Giles	108	105	213
St. Martin	110	108	218
St. Michael	112	110	222
St. Nicholas	115	112	227
St. Peter	118	115	233
St. Paul	120	118	238
St. James	122	120	242
St. John	125	122	247
St. Mary	128	125	253
St. Anne	130	128	258
St. Elizabeth	132	130	262
St. George	135	132	267
St. David	138	135	273
St. Thomas	140	138	278
St. Richard	142	140	282
St. Edmund	145	142	287
St. Andrew	148	145	293
St. Dunstons	150	148	298
St. Giles	152	150	302
St. Martin	155	152	307
St. Michael	158	155	313
St. Nicholas	160	158	318
St. Peter	162	160	322
St. Paul	165	162	327
St. James	168	165	333
St. John	170	168	338
St. Mary	172	170	342
St. Anne	175	172	347
St. Elizabeth	178	175	353
St. George	180	178	358
St. David	182	180	362
St. Thomas	185	182	367
St. Richard	188	185	373
St. Edmund	190	188	378
St. Andrew	192	190	382
St. Dunstons	195	192	387
St. Giles	198	195	393
St. Martin	200	198	398
St. Michael	202	200	402
St. Nicholas	205	202	407
St. Peter	208	205	413
St. Paul	210	208	418
St. James	212	210	422
St. John	215	212	427
St. Mary	218	215	433
St. Anne	220	218	438
St. Elizabeth	222	220	442
St. George	225	222	447
St. David	228	225	453
St. Thomas	230	228	458
St. Richard	232	230	462
St. Edmund	235	232	467
St. Andrew	238	235	473
St. Dunstons	240	238	478
St. Giles	242	240	482
St. Martin	245	242	487
St. Michael	248	245	493
St. Nicholas	250	248	498
St. Peter	252	250	502
St. Paul	255	252	507
St. James	258	255	513
St. John	260	258	518
St. Mary	262	260	522
St. Anne	265	262	527
St. Elizabeth	268	265	533
St. George	270	268	538
St. David	272	270	542
St. Thomas	275	272	547
St. Richard	278	275	553
St. Edmund	280	278	558
St. Andrew	282	280	562
St. Dunstons	285	282	567
St. Giles	288	285	573
St. Martin	290	288	578
St. Michael	292	290	582
St. Nicholas	295	292	587
St. Peter	298	295	593
St. Paul	300	298	598
St. James	302	300	602
St. John	305	302	607
St. Mary	308	305	613
St. Anne	310	308	618
St. Elizabeth	312	310	622
St. George	315	312	627
St. David	318	315	633
St. Thomas	320	318	638
St. Richard	322	320	642
St. Edmund	325	322	647
St. Andrew	328	325	653
St. Dunstons	330	328	658
St. Giles	332	330	662
St. Martin	335	332	667
St. Michael	338	335	673
St. Nicholas	340	338	678
St. Peter	342	340	682
St. Paul	345	342	687
St. James	348	345	693
St. John	350	348	698
St. Mary	352	350	702
St. Anne	355	352	707
St. Elizabeth	358	355	713
St. George	360	358	718
St. David	362	360	722
St. Thomas	365	362	727
St. Richard	368	365	733
St. Edmund	370	368	738
St. Andrew	372	370	742
St. Dunstons	375	372	747
St. Giles	378	375	753
St. Martin	380	378	758
St. Michael	382	380	762
St. Nicholas	385	382	767
St. Peter	388	385	773
St. Paul	390	388	778
St. James	392	390	782
St. John	395	392	787
St. Mary	398	395	793
St. Anne	400	398	798
St. Elizabeth	402	400	802
St. George	405	402	807
St. David	408	405	813
St. Thomas	410	408	818
St. Richard	412	410	822
St. Edmund	415	412	827
St. Andrew	418	415	833
St. Dunstons	420	418	838
St. Giles	422	420	842
St. Martin	425	422	847
St. Michael	428	425	853
St. Nicholas	430	428	858
St. Peter	432	430	862
St. Paul	435	432	867
St. James	438	435	873
St. John	440	438	878
St. Mary	442	440	882
St. Anne	445	442	887
St. Elizabeth	448	445	893
St. George	450	448	898
St. David	452	450	902
St. Thomas	455	452	907
St. Richard	458	455	913
St. Edmund	460	458	918
St. Andrew	462	460	922
St. Dunstons	465	462	927
St. Giles	468	465	933
St. Martin	470	468	938
St. Michael	472	470	942
St. Nicholas	475	472	947
St. Peter	478	475	953
St. Paul	480	478	958
St. James	482	480	962
St. John	485	482	967
St. Mary	488	485	973
St. Anne	490	488	978
St. Elizabeth	492	490	982
St. George	495	492	987
St. David	498	495	993
St. Thomas	500	498	998
St. Richard	502	500	1002
St. Edmund	505	502	1007
St. Andrew	508	505	1013
St. Dunstons	510	508	1018
St. Giles	512	510	1022
St. Martin	515	512	1027
St. Michael	518	515	1033
St. Nicholas	520	518	1038
St. Peter	522	520	1042
St. Paul	525	522	1047
St. James	528	525	1053
St. John	530	528	1058
St. Mary	532	530	1062
St. Anne	535	532	1067
St. Elizabeth	538	535	1073
St. George	540	538	1078
St. David	542	540	1082
St. Thomas	545	542	1087
St. Richard	548	545	1093
St. Edmund	550	548	1098
St. Andrew	552	550	1102
St. Dunstons	555	552	1107
St. Giles	558	555	1113
St. Martin	560	558	1118
St. Michael	562	560	1122
St. Nicholas	565	562	1127
St. Peter	568	565	1133
St. Paul	570	568	1138
St. James	572	570	1142
St. John	575	572	1147
St. Mary	578	575	1153
St. Anne	580	578	1158
St. Elizabeth	582	580	1162
St. George	585	582	1167
St. David	588	585	1173
St. Thomas	590	588	1178
St. Richard	592	590	1182
St. Edmund	595	592	1187
St. Andrew	598	595	1193
St. Dunstons	600	598	1198
St. Giles	602	600	1202
St. Martin	605	602	1207
St. Michael	608	605	1213
St. Nicholas	610	608	1218
St. Peter	612	610	1222
St. Paul	615	612	1227
St. James	618	615	1233
St. John	620	618	1238
St. Mary	622	620	1242
St. Anne	625	622	1247
St. Elizabeth	628	625	1253
St. George	630	628	1258
St. David	632	630	1262
St. Thomas	635	632	1267
St. Richard	638	635	1273
St. Edmund	640	638	1278
St. Andrew	642	640	1282
St. Dunstons	645	642	1287
St. Giles	648	645	1293
St. Martin	650	648	1298
St. Michael	652	650	1302
St. Nicholas	655	652	1307
St. Peter	658	655	1313
St. Paul	660	658	1318
St. James	662	660	1322
St. John	665	662	1327
St. Mary	668	665	1333
St. Anne	670	668	1338
St. Elizabeth	672	670	1342
St. George	675	672	1347
St. David	678	675	1353
St. Thomas	680	678	1358
St. Richard	682	680	1362
St. Edmund	685	682	1367
St. Andrew	688	685	1373
St. Dunstons	690	688	1378
St. Giles	692	690	1382
St. Martin	695	692	1387
St. Michael	698	695	1393
St. Nicholas	700	698	1398
St. Peter	702	700	1402
St. Paul	705	702	1407
St. James	708	705	1413
St. John	710	708	1418
St. Mary	712	710	1422
St. Anne	715	712	1427
St. Elizabeth	718	715	1433
St. George	720	718	1438
St. David	722	720	1442
St. Thomas	725	722	1447
St. Richard	728	725	1453
St. Edmund	730	728	1458
St. Andrew	732	730	1462
St. Dunstons	735	732	1467
St. Giles	738	735	1473
St. Martin	740	738	1478
St. Michael	742	740	1482
St. Nicholas	745	742	1487
St. Peter	748	745	1493
St. Paul	750	748	1498
St. James	752	750	1502
St. John	755	752	1507
St. Mary	758	755	1513
St. Anne	760	758	1518
St. Elizabeth	762	760	1522
St. George	765	762	1527
St. David	768	765	1533
St. Thomas	770	768	1538
St. Richard	772	770	1542
St. Edmund	775	772	1547
St. Andrew	778	775	1553
St. Dunstons	780	778	1558
St. Giles	782	780	1562
St. Martin	785	782	1567
St. Michael	788	785	1573
St. Nicholas	790	788	1578
St. Peter	792	790	1582
St. Paul	795	792	1587
St. James	798	795	1593
St. John	800	798	1598
St. Mary	802	800	1602
St. Anne	805	802	1607
St. Elizabeth	808	805	1613
St. George	810	808	1618
St. David	812	810	1622
St. Thomas	815	812	1627
St. Richard	818	815	1633
St. Edmund	820	818	1638
St. Andrew	822	820	1642
St. Dunstons	825	822	1647
St. Giles	828	825	1653
St. Martin	830	828	1658

mistakenly believing they spread the disease. Although the government took a much more active and organized role in trying to prevent the plague, their efforts ultimately failed because their actions were based on incorrect theories about bad air rather than exterminating the true culprits: the rats and fleas.

Q6. Evaluate the short-term significance of Harvey's discovery.

Pair & Share Activity

Q7. Why was William Harvey's brilliant discovery of blood circulation completely ignored in the prevention and treatment of the Great Plague of 1665?

Historian's Corner: The Debate over the Significance of William Harvey

Historians often debate whether William Harvey represents a sudden turning point or a very gradual stepping stone in medicine. Traditionalist historians celebrate his work as a revolutionary breakthrough that demolished Galenic physiology. However, revisionist historians emphasize the extreme limits of his immediate significance. They point out that a lot of contemporary doctors simply ignored or openly criticized Harvey because his discovery offered no practical treatment for sick patients. As a result, English medical textbooks continued to print Galen's flawed theories until 1651, and Harvey's findings were not taught in universities until 1673, decades after his discovery.

GCSE Exam Practice

Q8. Explain why the government's response to the Great Plague of 1665 was more effective than its response to the Black Death of 1348. (12 marks)

Q9. Explain one way in which the response to the Great Plague in London (1665) was different from the response to the Black Death (1348). (4 marks)